Understanding Itô Calculus

Álvaro Martín Ruiz

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1 Why a New Calculus Is Needed

Intuitive idea: In ordinary calculus, if we zoom into a smooth curve, each small piece looks linear. However, Brownian motion behaves differently: no matter how much we zoom in, the path remains jagged (we never get to "see" a straight line).

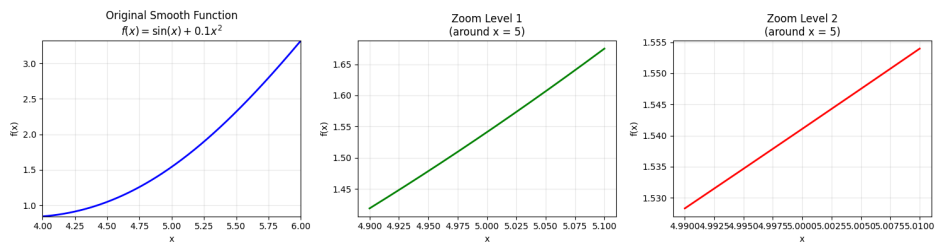

Figure 1: A smooth classical function appears locally linear under magnification.

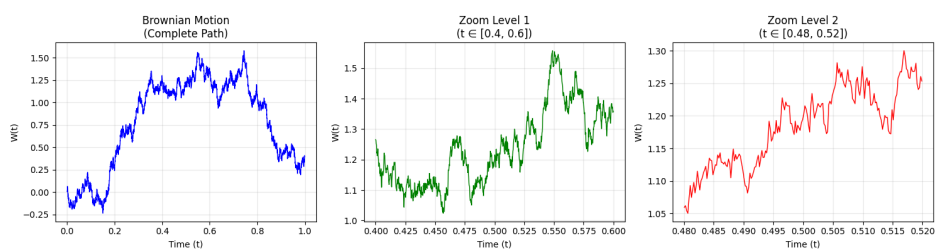

Figure 2: A Brownian path remains irregular at every scale.

Stochastic calculus extends ordinary calculus by handling this random, irregular behavior.

This is not just a technical annoyance. In many models (finance, physics, biology), the state variable we care about evolves under random noise modeled by Brownian motion. Therefore, we want to write and solve differential equations of the form

$$dX_t = \mu(X_t, t)\, dt + \sigma(X_t, t)\, dW_t$$

2 The Breakdown of Classical Calculus

2.1 The Riemann Integral

The standard Riemann integral,

$$\int_a^b f(t)dt$$

computes the accumulation of a quantity (typically the area under a curve) by dividing the time interval into small steps Δt, evaluating the function at a point within each step, and summing the results.

2.2 The Riemann–Stieltjes Integral

The Riemann-Stieltjes is a generalization of the Riemann integral. In the Riemann integral we add contributions proportional to how much space we cover, now to generalize we integrate a function $f(s)$ with respect to the changes in another arbitrary function $g(s)$.

$$\int_a^b f(x)dg(x)$$

If g(x) is x, we get the specific case of the Riemann Integral.

2.3 Why the Riemann–Stieltjes Integral Fails

In classical analysis, an integral of the form

$$\int_a^b f(s)\, dg(s)$$

is defined via Riemann–Stieltjes sums:

$$\sum f(t_i)\big(g(t_{i+1}) - g(t_i)\big).$$

where $a = t_0 < t_1 < \cdots < t_n = b$, are "small intervals" (partitions of the interval $[a, b]$).

The integral exists as long as:

- f is continuous on $[a, b]$.
- g is of bounded variation.

Because the total variation of a Brownian path on any interval is infinite, these sums fail to converge in the classical sense.

2.4 The Scaling Problem

2.4.1 Review of the Taylor Expansion

Let's review the Taylor expansion and how the finite difference formulas is derived.

Taylor's Theorem (Lagrange form of the remainder). Let f be $(n+1)$-times differentiable on an open interval containing a (center point where the derivatives are taken) and x (point where the function is evaluated). Then

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + R_n(x), \quad \text{where} \quad R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}(x-a)^{n+1}$$

for some ξ between a and x.

Therefore, if f is smooth, we can expand around the point x, meaning that x is the center point where the derivatives are taken, and $x+h$ is the nearby point where the function is evaluated:

$$f(x+h) = \sum_{n=0}^{\infty} \frac{h^n}{n!} f^{(n)}(x) = f(x) + hf'(x) + \frac{h^2}{2} f''(x) + \frac{h^3}{6} f^{(3)}(x) + \cdots .$$

In particular,

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2} f''(x) + O(h^3).$$

If we subtract $f(x)$ and divide by h:

$$\frac{f(x+h) - f(x)}{h} = f'(x) + \frac{h}{2} f''(x) + O(h^2)$$

Taking the limit as h becomes very small, the higher-order terms vanish, leaving us with:

$$\frac{f(x+h) - f(x)}{h} \approx f'(x) \text{ with an error of } O(h).$$

To translate this into the differential notation used in physics and engineering, we replace the small step h with the differential dX_t, and the change in the function $f(x+h) - f(x)$ with df. The expansion becomes:

$$df = f'(X_t)dX_t + \underbrace{\frac{1}{2} f''(X_t)(dX_t)^2 + O((dX_t)^3)}_{\text{Higher order terms}}$$

If X_t is a smooth, classical path, the $(dX_t)^2$ term shrinks to zero much faster than dX_t as the step size decreases. We throw the entire bracketed section away, leaving us with the standard chain rule: $df = f'(X_t)dX_t$.

However, when a function depends on Brownian motion, the second-order term in a Taylor expansion cannot be discarded. For Brownian increments, the scaling is different, and terms involving $(dB_t)^2$ contribute at leading order.

3 Quadratic Variation

Quadratic variation is the sum of the squared increments.

3.1 Definition of Quadratic Variation

Let

$$\Pi = \{0 = t_0 < t_1 < \cdots < t_n = t\}$$

be a partition of the interval $[0, t]$. Define the quadratic variation along this partition as

$$Q_\Pi(W) = \sum_{i=0}^{n-1} \big(W_{t_{i+1}} - W_{t_i}\big)^2.$$

We then refine the partition, letting the mesh size $\|\Pi\| \to 0$, and define

$$[W]_t = \lim_{\|\Pi\| \to 0} \sum_{i=0}^{n-1} \big(W_{t_{i+1}} - W_{t_i}\big)^2.$$

This limit, if it exists, is called the quadratic variation of the process.

3.2 Quadratic Variation of Brownian Motion

For Brownian motion, we have the fundamental identity:

$$[W]_t = t \text{ often written in differential form as } (dW_t)^2 = dt$$

This means that, for almost every realization of the Brownian path, the quadratic variation over the interval $[0, t]$ is exactly equal to t. It is also the fundamental reason the chain rule must be modified, second-order terms survive, and new integration theories are required.

Heuristic derivation.

To understand why this is true, consider a partition of the interval $[0, t]$ into N subintervals of equal size $\Delta t = \frac{t}{N}$. Denote the increments by:

$$\Delta W_i = W_{t_{i+1}} - W_{t_i}$$

By the properties of Brownian motion:

$$\Delta W_i \sim \mathcal{N}(0, \Delta t)$$

Recall that for a Gaussian random variable:

$$X \sim \mathcal{N}(0, \sigma^2) \quad \Rightarrow \quad \mathbb{E}[X^2] = \sigma^2$$

Thus, each increment satisfies:

$$\mathbb{E}[(\Delta W_i)^2] = \Delta t$$

Now consider the sum of squared increments:

$$\sum_{i=1}^{N} (\Delta W_i)^2$$

Taking expectations:

$$\mathbb{E}\left[\sum_{i=1}^{N} (\Delta W_i)^2\right] = \sum_{i=1}^{N} \mathbb{E}[(\Delta W_i)^2] = \sum_{i=1}^{N} \Delta t = t$$

This shows that, on average, the quadratic variation over $[0, t]$ equals the length of the interval.

Interpretation.

Although the increments themselves have mean zero, their squares accumulate in a perfectly regular way. This suggests that, while Brownian motion has no first-order structure, it has a well-defined second-order structure.

A more refined argument (L^2 convergence [1]).

The previous computation only shows that the expected value of the sum is t. To understand whether the sum itself converges, we must control its fluctuations.

Let $\Pi = \{0 = t_0 < t_1 < \cdots < t_n = t\}$ be a partition of $[0, t]$, and define:

$$Q_\Pi(W) = \sum_{i=1}^{n} (W_{t_i} - W_{t_{i-1}})^2$$

[1] The "L" stands for Lebesgue (a pioneer of integration) and the "2" refers to squared errors (variance). In probability, this is called *convergence in mean square*. Showing $\mathbb{E}[Q_\Pi(W)] = t$ just means the *average* shot hits the target, even if individual shots scatter wildly. Proving L^2 convergence guarantees that the expected squared error shrinks to zero, meaning the scatter completely vanishes and the approximation tightens perfectly onto t.

We study the mean squared deviation:

$$\mathbb{E}\big[(Q_\Pi(W) - t)^2\big]$$

Using the notation $\Delta W_i = W_{t_i} - W_{t_{i-1}}$, we write:

$$Q_\Pi(W) - t = \sum_{i=1}^{n} \big((\Delta W_i)^2 - (t_i - t_{i-1})\big)$$

Expanding the square and using independence of increments:

$$\begin{aligned}\mathbb{E}\big[(Q_\Pi(W) - t)^2\big] &= \sum_{i=1}^{n} \mathbb{E}\left[\big((\Delta W_i)^2 - (t_i - t_{i-1})\big)^2\right] \\ &= \sum_{i=1}^{n} \left(\mathbb{E}[(\Delta W_i)^4] - 2(t_i - t_{i-1})\mathbb{E}[(\Delta W_i)^2] + (t_i - t_{i-1})^2\right).\end{aligned}$$

For a Gaussian random variable $X \sim \mathcal{N}(0, \sigma^2)$, we have:

$$\mathbb{E}[X^2] = \sigma^2, \qquad \mathbb{E}[X^4] = 3\sigma^4$$

Applying this (recall that since $\Delta W_i \sim \mathcal{N}(0, \Delta t)$, its variance $\mathbb{E}[(\Delta W_i)^2]$ is exactly $(t_i - t_{i-1})$):

$$\mathbb{E}[(\Delta W_i)^4] = 3(t_i - t_{i-1})^2$$

Substituting:

$$\mathbb{E}\big[(Q_\Pi(W) - t)^2\big] = \sum_{i=1}^{n} \big(3(t_i - t_{i-1})^2 - 2(t_i - t_{i-1})^2 + (t_i - t_{i-1})^2\big) = 2\sum_{i=1}^{n}(t_i - t_{i-1})^2$$

Let $\|\Pi\| = \max_i(t_i - t_{i-1})$ denote the mesh size of the partition[2]. Then:

$$\sum_{i=1}^{n}(t_i - t_{i-1})^2 \leq \|\Pi\| \sum_{i=1}^{n}(t_i - t_{i-1}) = \|\Pi\| * [(t_1 - 0) + (t_2 - t_1) + (t_3 - t_2) + \cdots + (t - t_{n-1})] = \|\Pi\|\, t$$

To see why this inequality holds, we can expand the squared time step as $(t_i - t_{i-1})^2 = \Delta t_i \times \Delta t_i$. By definition, the mesh size is the largest interval in the partition, meaning every individual step is bounded by it: $\Delta t_i \leq \|\Pi\|$. If we replace one of the Δt_i factors with this maximum value, we

[2]We define the limit using the maximum step size ($\|\Pi\| \to 0$) instead of letting the number of partitions go to infinity ($n \to \infty$), because if we only required $n \to \infty$, we could chop one segment of the time interval into an infinite number of small pieces while leaving another segment as a single chunk. By forcing the *largest* individual time step to shrink to zero, we mathematically guarantee that every single part of the interval is refined infinitely finely.

establish the inequality. Because $\|\Pi\|$ is a constant maximum for the given partition, we can factor it entirely outside the summation. What remains inside the sum is simply the accumulation of all the consecutive time steps, which perfectly reconstructs the total time t.

Thus:

$$\mathbb{E}\big[(Q_\Pi(W) - t)^2\big] \leq 2\,\|\Pi\|\,t$$

As the partition is refined, $\|\Pi\| \to 0$, and therefore:

$$\lim_{\|\Pi\|\to 0} \mathbb{E}\big[(Q_\Pi(W) - t)^2\big] = 0$$

This shows that the quadratic variation converges to t in L^2, and therefore also in probability.

Conclusion.

This result reveals a fundamental structural property of Brownian motion:

- First-order variation (absolute increments) diverges,
- but second-order variation (squared increments) converges to a deterministic quantity.

This is the key reason why quadratic variation replaces total variation as the correct notion of accumulated fluctuation in stochastic calculus.

A new notion of accumulated variation.

In classical calculus, variation is measured through absolute increments, leading to finite total variation for regular paths. For Brownian motion, this approach fails.

Quadratic variation provides an alternative:

$$\text{Total variation: } \sum |\Delta W| \to \infty \qquad \text{Quadratic variation: } \sum (\Delta W)^2 \to t$$

This suggests that the correct way to measure the evolution of a Brownian path is not through its absolute displacement, but through the accumulation of its squared fluctuations.

4 The Itô Integral

The previous sections showed that classical integration fails for Brownian motion due to its infinite variation. However, we also identified a key structural property that will allow us to move forward: finite quadratic variation.

This property is one of the key ingredients that makes stochastic calculus possible. The goal is to give a rigorous meaning to integrals of the form:

$$\int_0^t H_s \, dW_s$$

where H_s is a process that may depend on time and on the past behavior of W.

Unlike classical integrals, this expression cannot be defined pathwise. Instead, we must construct it carefully as a limit of discrete approximations.

4.1 Step 1: Riemann-Type Sums

We begin by mimicking the classical construction.

Let $\Pi = \{0 = t_0 < t_1 < \cdots < t_n = t\}$ be a partition of $[0, t]$. Consider the sum:

$$\sum_{i=0}^{n-1} H_{t_i} \big(W_{t_{i+1}} - W_{t_i}\big)$$

This can be interpreted as accumulating the impact of random increments ΔW_i, each weighted by the value of H at the beginning of the interval.

Notice that this is very similar to the classical Riemann–Stieltjes integral, but with one crucial difference: the **sampling point**[3].

In the classical Riemann-Stieltjes integral, the sampling point inside each interval is arbitrary. For example, all of the following are valid approximations:

$$\sum f(t_i)\big(g(t_{i+1}) - g(t_i)\big), \quad \sum f(t_{i+1})\big(g(t_{i+1}) - g(t_i)\big), \quad \sum f\left(\frac{t_{i+1} + t_i}{2}\right)\big(g(t_{i+1}) - g(t_i)\big)$$

As the partition is refined, all these sums converge to the same limit when g has bounded variation.

[3] **Sampling point:** The point chosen within a subinterval $[t_i, t_{i+1}]$ at which the integrand is evaluated. In Riemann-type sums, it determines the "height" of the rectangle used to approximate the contribution of that subinterval.

However, in the stochastic setting, this is no longer true.

Because Brownian motion has infinite variation, the limit of these sums may fail to exist or may depend on the choice of sampling point. Different choices lead to different limiting objects. Therefore, for the Itô integral, we make a specific choice:

- The integrand is always evaluated at the **left endpoint**.

Why the left endpoint?

This choice ensures that H_{t_i} depends only on information available up to time t_i, and not on future increments of the Brownian motion.[4]

4.2 Step 2: Taking the Limit

We now take the limit as the partition is refined:

$$\int_0^t H_s \, dW_s = \lim_{\|\Pi\| \to 0} \sum_{i=0}^{n-1} H_{t_i} (W_{t_{i+1}} - W_{t_i})$$

Unlike classical integration, this limit is not defined pathwise[5]. Instead, it is defined in a probabilistic sense (typically in L^2).

- **Explanation:** In classical integration, we fix a function (a path) and compute the limit of Riemann sums along that function. When the integrator has bounded variation, this limit exists and is independent of the choice of sampling points.

  However, for Brownian motion, each realization $W_t(\omega)$ is highly irregular (it has infinite variation). As a result, even for a fixed path, the Riemann-type sums may fail to converge, or may converge to different values depending on how the sampling points are chosen.

  Therefore, unlike in classical integration—where convergence holds path by path—we define the stochastic integral through convergence in a probabilistic sense. This means that the approximating sums converge "on average" (typically in L^2), rather than for each individual path.

[4] This property is called *non-anticipativity* or *adaptedness*. It ensures that the process H_t cannot "see the future" of W_t. In stochastic modeling this corresponds to the natural requirement that decisions can only depend on currently available information.

[5] A path is a single realization (or outcome) of a stochastic process, i.e., one possible trajectory $t \mapsto W_t(\omega)$.

4.3 Key Property: The Itô Isometry

One of the key results that makes the Itô integral well-behaved is the following identity:

$$\mathbb{E}\left[\left(\int_0^t H_s\, dW_s\right)^2\right] = \mathbb{E}\left[\int_0^t H_s^2\, ds\right]$$

This is known as the **Itô isometry**:

- On the left-hand side, we have a stochastic integral involving Brownian motion.
- On the right-hand side, we have an ordinary integral in time.

The isometry tells us that, in L^2, integrating against Brownian motion behaves in a very controlled way: the "size" of the stochastic integral is exactly determined by the size of the integrand.

Proof (for simple processes).

We begin by proving the result for a simple process of the form [6]

$$H_s = \sum_{i=0}^{n-1} H_{t_i}\, \mathbf{1}_{[t_i, t_{i+1})}(s),$$

where each H_{t_i} is $\mathcal{F}_{t_i}$-measurable.

In this case, as we have seen before, the Itô integral reduces to:

$$\int_0^t H_s\, dW_s = \sum_{i=0}^{n-1} H_{t_i}(W_{t_{i+1}} - W_{t_i})$$

We now compute the second moment:

$$\mathbb{E}\left[\left(\int_0^t H_s\, dW_s\right)^2\right] = \mathbb{E}\left[\left(\sum_{i=0}^{n-1} H_{t_i}\Delta W_i\right)^2\right]$$

Expanding the square:

$$\mathbb{E}\left[\left(\sum_{i=0}^{n-1} H_{t_i}\Delta W_i\right)^2\right] = \mathbb{E}\left[\sum_{i,j} H_{t_i} H_{t_j}\Delta W_i \Delta W_j\right]$$

We now split the sum into two cases: $i \neq j$ and $i = j$.

[6] The indicator function $\mathbf{1}_A(x)$ equals 1 if $x \in A$ and 0 otherwise.

- **Cross terms ($i \neq j$):** These terms vanish.By symmetry, we can assume without loss of generality that $i < j$. We have
$$\mathbb{E}\left[H_{t_i} H_{t_j}\, \Delta W_i\, \Delta W_j\right]$$
where:
  - $H_{t_i} H_{t_j}\, \Delta W_i$ depends only on information up to time t_j.
  - $\Delta W_j = W_{t_{j+1}} - W_{t_j}$ is a future increment strictly after t_j.
  - Brownian increments are independent[7] of the past and have mean zero.

  As a result,
$$\mathbb{E}[H_{t_i} H_{t_j} \Delta W_i \Delta W_j] = \mathbb{E}\left[H_{t_i} H_{t_j}\, \Delta W_i\, \underbrace{\mathbb{E}\left[\Delta W_j \mid \mathcal{F}_{t_j}\right]}_{0}\right] = 0 \quad \text{for } i \neq j.$$

- **Diagonal terms ($i = j$):** These remain.

  In this case,
$$\mathbb{E}[H_{t_i}^2 (\Delta W_i)^2] = \mathbb{E}\left[H_{t_i}^2\, \mathbb{E}[(\Delta W_i)^2 \mid \mathcal{F}_{t_i}]\right] = \mathbb{E}[H_{t_i}^2 (t_{i+1} - t_i)].$$

Putting everything together:
$$\mathbb{E}\left[\left(\int_0^t H_s\, dW_s\right)^2\right] = \sum_{i=0}^{n-1} \mathbb{E}[H_{t_i}^2 (t_{i+1} - t_i)]$$

Rewriting:
$$= \mathbb{E}\left[\sum_{i=0}^{n-1} H_{t_i}^2 (t_{i+1} - t_i)\right] = \mathbb{E}\left[\int_0^t H_s^2\, ds\right]$$

This step follows from the definition of the integral for simple processes. Since H_s is constant on each interval $[t_i, t_{i+1})$, we have
$$\int_0^t H_s^2\, ds = \sum_{i=0}^{n-1} H_{t_i}^2 (t_{i+1} - t_i)$$

This completes the proof for simple processes. The general case follows by approximation.

[7] Recall that the expectation of the product of two independent variables X and Y is the product of their expectations: $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$.

4.4 Key Property: Martingale Structure

Another fundamental property is:

$$\mathbb{E}\left[\int_0^t H_s\, dW_s \,\middle|\, \mathcal{F}_s\right] = \int_0^s H_u\, dW_u \qquad s \le t$$

This means that the Itô integral is a martingale [8].The integral has no predictable drift and it evolves purely through accumulated randomness.

Heuristic proof

We argue for simple processes. In this case,

$$\int_0^t H_u\, dW_u = \sum_{i=0}^{n-1} H_{t_i}\big(W_{t_{i+1}} - W_{t_i}\big)$$

Fix $s < t$, and suppose $s = t_k$ is one of the partition points (for simplicity[9]).

We split the sum into two parts:

$$\int_0^t H_u\, dW_u = \underbrace{\sum_{i=0}^{k-1} H_{t_i}\big(W_{t_{i+1}} - W_{t_i}\big)}_{\text{past}} + \underbrace{\sum_{i=k}^{n-1} H_{t_i}\big(W_{t_{i+1}} - W_{t_i}\big)}_{\text{future}}$$

Now take conditional expectation with respect to $\mathcal{F}_s$.

- **Past part:** All terms in the first sum involve only times up to s, so they are $\mathcal{F}_s$-measurable [10]. Therefore,
$$\mathbb{E}\left[\sum_{i=0}^{k-1} H_{t_i}\big(W_{t_{i+1}} - W_{t_i}\big) \,\middle|\, \mathcal{F}_s\right] = \sum_{i=0}^{k-1} H_{t_i}\big(W_{t_{i+1}} - W_{t_i}\big).$$
- **Future part:** For $i \ge k$, the increment $W_{t_{i+1}} - W_{t_i}$ is independent of $\mathcal{F}_s$ and has mean zero (for Brownian motion, $\mathbb{E}[W_{t_{i+1}} - W_{t_i}] = 0$). Using the Tower Property [11] of conditional expectation (since $s \le t_i$, we have $\mathcal{F}_s \subset \mathcal{F}_{t_i}$):
$$\mathbb{E}\left[H_{t_i}\big(W_{t_{i+1}} - W_{t_i}\big) \mid \mathcal{F}_s\right] = \mathbb{E}\left[\mathbb{E}\left[H_{t_i}\big(W_{t_{i+1}} - W_{t_i}\big) \mid \mathcal{F}_{t_i}\right] \mid \mathcal{F}_s\right]$$

[8] A stochastic process $(M_t)_{t\ge 0}$ is a martingale with respect to a filtration $(\mathcal{F}_t)$ if $\mathbb{E}[|M_t|] < \infty$ and $\mathbb{E}[M_t \mid \mathcal{F}_s] = M_s$ for all $s \le t$. Informally, it is a fair game that satisfies $\mathbb{E}[M_{t+1} \mid \mathcal{F}_t] = M_t$, meaning that given everything known to now, the best prediction of the future is the present value.

[9] For a more detailed proof, see *Stochastic Calculus for Finance II: Continuous-Time Models* by Steven E. Shreve, Chapter 4.2.2.

[10] Being $\mathcal{F}$-measurable means that the value of X can be determined using only the information contained in $\mathcal{F}$.

[11] The Tower Property states that if we have nested information sets (such as $\mathcal{F}_s \subset \mathcal{F}_{t_i}$ for $s \le t_i$) conditioning on the smaller (older) information set overrides the larger (newer) one: $\mathbb{E}[\mathbb{E}[X \mid \mathcal{F}_{t_i}] \mid \mathcal{F}_s] = \mathbb{E}[X \mid \mathcal{F}_s]$. Intuitively, your best estimate today of what your best estimate will be tomorrow is simply your best estimate today.

$$= \mathbb{E}\left[H_{t_i} \underbrace{\mathbb{E}\left[W_{t_{i+1}} - W_{t_i} \mid \mathcal{F}_{t_i}\right]}_{0} \,\middle|\, \mathcal{F}_s \right] = 0$$

and all terms in the second sum vanish.

Putting everything together:

$$\mathbb{E}\left[\int_0^t H_u \, dW_u \,\middle|\, \mathcal{F}_s \right] = \sum_{i=0}^{k-1} H_{t_i}\big(W_{t_{i+1}} - W_{t_i}\big) = \int_0^s H_u \, dW_u.$$

5 Itô's Formula

5.1 Why the Classical Chain Rule Fails

As a brief recap of the main concepts that we saw in section 2:

1. In classical calculus, the chain rule follows from the Taylor expansion:

$$df = f'(X_t)dX_t + \frac{1}{2}f''(X_t)(dX_t)^2 + \cdots$$

2. For smooth processes, the second-order term vanishes, because $(dX_t)^2 \ll dX_t$ and we recover:

$$df = f'(X_t)dX_t$$

3. However, for Brownian motion, the scaling is different: $dW_t \sim \sqrt{dt}$ and $(dW_t)^2 \sim dt$, which means that the second-order term does not vanish (in fact, it contributes at the same order as the first-order term).

5.2 Statement of Itô's Formula

The correct version of the chain rule is given by Itô's formula.

For a twice differentiable function f, we have:

$$df(W_t) = f'(W_t)\, dW_t + \frac{1}{2}f''(W_t)\, dt.$$

In integral form:

$$f(W_t) = f(W_0) + \int_0^t f'(W_s)\, dW_s + \frac{1}{2}\int_0^t f''(W_s)\, ds.$$

The formula has two components:

- A stochastic term involving dW_t
- A correction term that accumulates over time (proportional to dt)

This additional dt term is a direct consequence of the quadratic variation of Brownian motion[12].

5.3 Heuristic Derivation

Starting from the Taylor expansion:

$$f(W_{t+\Delta t}) - f(W_t) = f'(W_t)\Delta W + \frac{1}{2}f''(W_t)(\Delta W)^2 + O((\Delta W)^3),$$

we use the key scaling properties of Brownian motion:

$$(\Delta W)^2 \approx \Delta t, \qquad (\Delta W)^3 \approx 0.$$

Substituting these into the expansion:

$$f(W_{t+\Delta t}) - f(W_t) \approx f'(W_t)\Delta W + \frac{1}{2}f''(W_t)\Delta t.$$

Passing to differential notation, this becomes:

$$df(W_t) = f'(W_t)\, dW_t + \frac{1}{2}f''(W_t)\, dt.$$

5.4 Examples

5.4.1 $f(x) = x^2$

Let us apply Itô's formula to the function $f(x) = x^2$.

We compute:

$$f'(x) = 2x, \quad f''(x) = 2.$$

Applying Itô's formula:

$$d(W_t^2) = 2W_t\, dW_t + dt.$$

Integrating:

$$W_t^2 = W_0^2 + \int_0^t 2W_s\, dW_s + t.$$

[12] $(dW_t)^2 \sim dt$

5.4.2 $dS_t =$ GBM and $f(x) = \log(x)$

Let S_t follow a geometric Brownian motion:

$$dS_t = \mu S_t \, dt + \sigma S_t \, dW_t.$$

We apply Itô's formula to the function $f(x) = \log(x)$.

We compute:

$$f'(x) = \frac{1}{x}, \quad f''(x) = -\frac{1}{x^2}.$$

Applying Itô's formula:

$$d(\log S_t) = \frac{1}{S_t} dS_t + \frac{1}{2}\left(-\frac{1}{S_t^2}\right)(dS_t)^2.$$

We now substitute dS_t:

$$\frac{1}{S_t} dS_t = \mu dt + \sigma dW_t.$$

Using the rule $(dW_t)^2 = dt$, we have:

$$(dS_t)^2 = \sigma^2 S_t^2 dt.$$

Therefore:

$$\frac{1}{2}\left(-\frac{1}{S_t^2}\right)(dS_t)^2 = -\frac{1}{2}\sigma^2 dt.$$

Putting everything together:

$$d(\log S_t) = \left(\mu - \frac{1}{2}\sigma^2\right) dt + \sigma dW_t$$

And if we want to solve for S_t:

$$\begin{aligned}
\int_0^t d(\log S_u) &= \int_0^t \left(\mu - \frac{1}{2}\sigma^2\right) du + \int_0^t \sigma \, dW_u \\
\log S_t - \log S_0 &= \left(\mu - \frac{1}{2}\sigma^2\right) t + \sigma W_t \\
\log S_t &= \log S_0 + \left(\mu - \frac{1}{2}\sigma^2\right) t + \sigma W_t \\
S_t &= e^{\log S_0 + \left(\mu - \frac{1}{2}\sigma^2\right) t + \sigma W_t} \\
S_t &= S_0 e^{\left(\mu - \frac{1}{2}\sigma^2\right) t + \sigma W_t}
\end{aligned}$$

As a clarification, the stochastic integral in the first step evaluates to:

$$\int_0^t \sigma \, dW_u = \sigma(W_t - W_0) = \sigma W_t,$$

since $W_0 = 0$.

Also, note that the final expression represents the analytical solution for the price of a stock at time t following a Geometric Brownian Motion (GBM).

5.4.3 Itô's formula for evaluating an integral

One useful application of Itô's formula is to compute stochastic integrals.

Consider again the function $f(x) = x^2$. From the previous result:

$$d(W_t^2) = 2W_t \, dW_t + dt.$$

Rearranging:

$$2W_t \, dW_t = d(W_t^2) - dt.$$

Integrating both sides from 0 to t:

$$\int_0^t 2W_s \, dW_s = W_t^2 - W_0^2 - t.$$

Dividing by 2:

$$\int_0^t W_s \, dW_s = \frac{1}{2}(W_t^2 - t).$$

This provides an explicit expression for the stochastic integral.

In contrast to classical calculus, such integrals cannot be computed directly from the definition. Itô's formula provides a way to evaluate them indirectly.

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